

Task 1:

Creation of VLANs (Sales & Support) on Cisco Switch

Objective: Access the Cisco IOS Command Line Interface (CLI) on a Layer 2 switch and configure two distinct broadcast domains: VLAN 10 named 'Sales' and VLAN 20 named 'Support'.

1. Cisco IOS CLI Commands Executed:

```
Switch> enable
```

```
Switch# configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
! Creating VLAN 10 (Sales)
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# exit
```

```
! Creating VLAN 20 (Support)
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Support
Switch(config-vlan)# exit
Switch(config)# end
Switch# copy running-config startup-config
```

2. Verification Command & CLI Output:

To verify the creation of VLAN 10 and VLAN 20 on the switch interface, execute show vlan brief:

 | **Switch# show vlan brief**

VLAN Information on Cisco Switch

Switch# show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	Sales	active	
20	Support	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

What This Output Means?

VLAN	Name	Status	Ports
1	default	active	All 24 switch ports (Fa0/1 – Fa0/24)
10	Sales	active	No ports assigned (used for specific devices)
20	Support	active	No ports assigned (used for specific devices)
1002	fddi-default	act/unsup	System VLAN
1003	token-ring-default	act/unsup	System VLAN
1004	fddinet-default	act/unsup	System VLAN
1005	trnet-default	act/unsup	System VLAN

 **Key Points**

- ✓ VLAN 1 (default) is active and has all 24 ports.
- ✓ VLAN 10 (Sales) and VLAN 20 (Support) are active but no ports are assigned yet.
- ✓ VLAN 1002–1005 are system VLANs (not user-defined).
- ✓ Status 'act/unsup' means the VLAN is active but not supported (system use).

 **VLAN Types**

- 1** Default VLAN (native VLAN)
- 10** User-defined VLAN (Sales)
- 20** User-defined VLAN (Support)
- 1002–1005** System VLANs (FDDI, Token Ring, FDDI-Net, TR-Net)

 **Summary**

This output shows the VLAN configuration on the Cisco switch, including:

- VLAN IDs and names
- Status of each VLAN
- Ports assigned to each VLAN
- System VLANs (1002–1005)

Port Range	Applied Mode	Assigned VLAN ID	VLAN Name	Status
FastEthernet 0/1 - 0/3	Access Port	VLAN 10	Sales	Active & Assigned
FastEthernet 0/4 - 0/5	Access Port	VLAN 20	Support	Active & Assigned

Task 2:

Assigning Switch Ports to VLANs & Verification

Objective: Use the Cisco IOS interface range command to efficiently assign switch ports Fa0/1 - Fa0/3 to VLAN 10 (Sales) and Fa0/4 - Fa0/5 to VLAN 20 (Support), then verify membership.

1. Cisco IOS CLI Commands Executed:

```
Switch> enable
```

```
Switch# configure terminal
```

```
! Assigning 3 Ports (Fa0/1 to Fa0/3) to VLAN 10 (Sales)
```

```
Switch(config)# interface range fastEthernet 0/1 - 3
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit
```

```
! Assigning 2 Ports (Fa0/4 to Fa0/5) to VLAN 20 (Support)
```

```
Switch(config)# interface range fastEthernet 0/4 - 5
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# end
```

```
Switch# copy running-config startup-config
```

2. Verification Command & Output:

To verify the updated VLAN membership and interface mapping, execute show vlan brief:

```
Switch# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

2. Port Assignment Summary Table:

Port Range	Applied Mode	Assigned VLAN ID	VLAN Name	Status
FastEthernet 0/1 - 0/3	Access Port	VLAN 10	Sales	Active & Assigned
FastEthernet 0/4 - 0/5	Access Port	VLAN 20	Support	Active & Assigned

Task 2:

Assigning Switch Ports to VLANs & Verification

Objective: Use the Cisco IOS interface range command to efficiently assign switch ports Fa0/1 - Fa0/3 to VLAN 10 (Sales) and Fa0/4 - Fa0/5 to VLAN 20 (Support), then verify membership.

1. Cisco IOS CLI Commands Executed:

```
Switch> enable
```

```
Switch# configure terminal
```

```
! Assigning 3 Ports (Fa0/1 to Fa0/3) to VLAN 10 (Sales)
```

```
Switch(config)# interface range fastEthernet 0/1 - 3
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit
```

```
! Assigning 2 Ports (Fa0/4 to Fa0/5) to VLAN 20 (Support)
```

```
Switch(config)# interface range fastEthernet 0/4 - 5
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# end
```

```
Switch# copy running-config startup-config
```

2. Verification Command & Output:

To verify the updated VLAN membership and interface mapping, execute show vlan brief:

```
Switch# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5
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Port Range	Applied Mode	Assigned VLAN ID	VLAN Name	Status
FastEthernet 0/1 - 0/3	Access Port	VLAN 10	Sales	Active & Assigned
FastEthernet 0/4 - 0/5	Access Port	VLAN 20	Support	Active & Assigned

Task 3:

Inter-Router Static Routing Configuration

Objective: Establish a point-to-point IP connection between two routers, configure subnets on WAN and LAN interfaces, and implement a static route on Router 1 to reach Router 2's internal LAN segment.

1. Network Addressing Scheme & Topology Table:

Router Name	Interface Port	Assigned IP Address	Subnet Mask	Connected Network Segment
Router 1	GigabitEthernet0/0/0	10.0.0.1	255.255.255.252	Inter-Router WAN Link
Router 1	GigabitEthernet0/0/1	192.168.1.1	255.255.255.0	Router 1 LAN (192.168.1.0/24)

Rou ter Na me	Interface Port	Assign ed IP Addre ss	Subnet Mask	Connect ed Network Segment
Rou ter 2	GigabitEther net0/0/0	10.0.0. 2	255.255.2 55.252	Inter- Router WAN Link
Rou ter 2	GigabitEther net0/0/1	192.16 8.2.1	255.255.2 55.0	Router 2 LAN (192.168. 2.0/24)

2. Router Configurations (Cisco IOS CLI):

Router 1 Interface & Static Route Setup:

Router1> enable

Router1# configure terminal

! Configure WAN Interface connecting to Router 2

Router1(config)# interface GigabitEthernet0/0/0

Router1(config-if)# ip address 10.0.0.1 255.255.255.252

```
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
```

! Configure Local LAN Interface

```
Router1(config)# interface GigabitEthernet0/0/1
```

```
Router1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
```

! Configure Static Route to reach LAN behind Router 2 via
Next-Hop IP 10.0.0.2

```
Router1(config)# ip route 192.168.2.0 255.255.255.0  
10.0.0.2
```

```
Router1(config)# end
```

```
Router1# copy running-config startup-config
```

Router 2 Interface Setup:

```
Router2> enable
```

```
Router2# configure terminal
```

! Configure WAN Interface connecting to Router 1

```
Router2(config)# interface GigabitEthernet0/0/0
```

```
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
```

```
Router2(config-if)# no shutdown
```

```
Router2(config-if)# exit
```

! Configure Local LAN Interface behind Router 2

```
Router2(config)# interface GigabitEthernet0/0/1
```

```
Router2(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
Router2(config-if)# no shutdown
```

```
Router2(config-if)# exit
```

```
Router2(config)# end
```

```
Router2# copy running-config startup-config
```

3. Routing Table Verification on Router 1:

To verify the configured static route in Router 1's routing table, execute show ip route:

Router1# show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets

C 10.0.0.0 is directly connected, GigabitEthernet0/0/0

192.168.1.0/24 is subnetted, 1 subnets

C 192.168.1.0 is directly connected,
GigabitEthernet0/0/1

192.168.2.0/24 is subnetted, 1 subnets

S 192.168.2.0 [1/0] via 10.0.0.2

Task 4:

Inter-VLAN Communication Troubleshooting & ROAS Solution

Objective: Simulate an inter-VLAN traffic failure scenario between Sales (VLAN 10) and Support (VLAN 20), analyze why Layer 2 traffic isolation causes ping failure, and provide the exact Router-on-a-Stick (ROAS) configuration to enable inter-VLAN routing.

1. Simulation Scenario & Test Result:

- **Source Device:** PC-Sales (IP: 192.168.10.10/24, Subnet: 255.255.255.0, VLAN 10)
 - **Destination Device:** PC-Support (IP: 192.168.20.10/24, Subnet: 255.255.255.0, VLAN 20)
 - **Command Executed:** ping 192.168.20.10
 - **Test Result: 100% Packet Loss (Request Timed Out)**
-

2. Why the Ping Fails (Technical Analysis):

1. **Layer 2 Broadcast Isolation:** VLANs logically divide a single physical switch into isolated Layer 2 broadcast domains. By design, switches never forward Ethernet

frames directly between different VLAN IDs without a Layer 3 routing engine.

- 2. Different IP Subnets:** 192.168.10.0/24 and 192.168.20.0/24 belong to distinct Layer 3 IP subnets. A Layer 2 switch lacks routing capabilities to rewrite MAC address headers and pass packets across different subnets.

3. Solution: Router-on-a-Stick (ROAS) Configuration:

To allow inter-VLAN communication, we connect a router interface to the switch via an 802.1Q Trunk link and configure sub-interfaces with default gateways for each VLAN.

Step A: Configure Switch Trunk Port

Code snippet

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# interface FastEthernet 0/24
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# switchport trunk allowed vlan 10,20
```

```
Switch(config-if)# exit
```


Step B: Configure Router Sub-Interfaces (Router-on-a-Stick)

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface GigabitEthernet 0/0/0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

! Configure Sub-interface for VLAN 10 (Sales Gateway)

```
Router(config)# interface GigabitEthernet 0/0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1  
255.255.255.0
```

```
Router(config-subif)# exit
```

! Configure Sub-interface for VLAN 20 (Support Gateway)

```
Router(config)# interface GigabitEthernet 0/0/0.20
```

```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.20.1  
255.255.255.0
```

Router(config-subif)# exit

4. Post-Configuration Result:

Once default gateways (192.168.10.1 and 192.168.20.1) are assigned to the respective PCs, the router successfully routes Layer 3 packets between VLAN 10 and VLAN 20, achieving 0% **Packet Loss (Ping Successful)**.